

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9.	(a)			Any five of following for (1) each - Alkenes contain σ and π bonds σ bonds are formed from s-s orbital overlap / end-on orbital overlap π bonds are formed from sideways overlap of p orbitals / overlap above and below plane - The π bond gives a region of high electron density - This is susceptible to electrophilic attack/ attack by an electron deficient species (This attack) leads to addition reactions	5			5		
	(b)			Diagram to show Correct dipole on Br ₂ (1) Two correct arrows (1) Formula of intermediate and arrow from lone pair or negative charge (1) Correct product (1)	4			4		
				Question 9 total	9	0	0	9	0	0